The following table summarizes GM changes by segment for fiscal years 2019 and 2018:

| | 2019 | | 2018 | | Increase (Decrease) | |
|---|---|---|---|---|---|---|
| | (Dollar amounts in thousands) | | | | | |
| | GM$ | GM% | GM$ | GM% | GM$ | GM% |
| TS | $ 13,889 | 20 % | $ 12,262 | 20 % | $ 1,627 | — % |
| HPP | 4,137 | 52 % | 6,137 | 59 % | (2,000) | (7)% |
| Total | $ 18,026 | 23 % | $ 18,399 | 25 % | $ (373) | (2)% |

The impact of product mix within our TS segment on gross margins for the fiscal years ended September 30 was as follows:

| | 2019 | | 2018 | | Increase | |
|---|---|---|---|---|---|---|
| | GM$ | GM% | GM$ | GM% | GM$ | GM% |
| | (Dollar amounts in thousands) | | | | | |
| Products | $ 7,462 | 13 % | $ 6,886 | 13 % | $ 576 | — % |
| Services | 6,427 | 56 % | 5,376 | 55 % | 1,051 | 1 % |
| Total | $ 13,889 | 20 % | $ 12,262 | 20 % | $ 1,627 | — % |

The overall TS segment gross margin as a percentage of sales remained the same in fiscal year 2019 when compared to fiscal year 2018. The $0.6 million increase in our TS segment product gross margins resulted from an increase in product revenues in the U.S. division, partially offset by a decrease in the U.K division. The $1.1 million increase in the TS segment service gross margins primarily resulted from increased service revenues in the U.S. division.

The impact of product mix on gross margins within our HPP segment for the fiscal years ended September 30 was as follows:

| | 2019 | | 2018 | | Increase (Decrease) | |
|---|---|---|---|---|---|---|
| | (Dollar amounts in thousands) | | | | | |
| | GM$ | GM% | GM$ | GM% | GM$ | GM% |
| Products | $ 2,719 | 42 % | $ 2,775 | 40 % | $ (56) | 2 % |
| Services | 1,418 | 95 % | 3,362 | 97 % | (1,944) | (2)% |
| Total | $ 4,137 | 52 % | $ 6,137 | 59 % | $ (2,000) | (7)% |

The overall HPP segment gross margins as a percentage of sales decreased to 52% in fiscal year 2019 from 59% in fiscal year 2018. The 7% decrease in gross margin as a percentage of sales in the HPP segment was primarily attributed to the impact of a decrease of $1.7 million in high margin Multicomputer royalty revenues.

***Engineering and Development Expenses***

The following table details our engineering and development expenses by operating segment for the fiscal years ended September 30, 2019 and 2018:

| | For the Year ended September 30, | | | | | |
|---|---|---|---|---|---|---|
| (Dollar amounts in thousands) | 2019 | % of Total | 2018 | % of Total | $ Decrease | % Decrease |
| | (Dollar amounts in thousands) | | | | | |
| **By Operating Segment:** | | | | | | |
| TS | $ — | — % | $ — | — % | $ — | — % |
| HPP | 2,800 | 100 % | 3,277 | 100 % | (477) | (15)% |
| Total | $ 2,800 | 100 % | $ 3,277 | 100 % | $ (477) | (15)% |

Engineering and development expenses decreased by $0.5 million to $2.8 million for the fiscal year ended September 30, 2019 as compared to $3.3 million for the fiscal year ended September 30, 2018. The current fiscal year expenses were primarily for product engineering expenses incurred in connection with the development of the new